AIRCRAFT FLIGHT INSTRUMENTS AND NAVIGATION SOURCES

FULFILLS PA.VI.B, CA.VI.B, AI.II.H

Objective	
The student shall understand ground and satellite based navigation, radar services from ATC, and the functions and capabilities of transponders and ADSB. The student shall become familiar with utilizing these functions on local and cross country flights.	
Instructor Actions	**Student Actions**
- Explain ground-based navigation aids - Demonstrate ERAU VOR simulator - Explain satellite-based navigational systems - Demonstrate Garmin GPS 175 simulator - Discuss ATC radar services and flight following - Discuss functions and capabilities of ADSB and associated systems - Walk around aircraft and identify antennas and their function	- Take notes and participate in instructor's discussion - Practice self-quizzing and verifying expected depiction on VOR simulator - Read GPS 175 pilot's guide and practice using iPad simulator - Understand limitations of navigation systems
Case Studies	**Equipment**
- Non-broadcasted traffic	- Additional resources - Computer/iPad - FAR/AIM - PHAK - White Board
Completion Standards	
The student shall explain identification and mitigation methods for aeromedical phenomena. The student shall demonstrate proficiency in incorporating navigation systems and radar services into local and cross country flights.	

ELEMENTS

RESOURCES

FAA-S-ACS-6C Private Pilot ACS - Area I Task H
FAA-S-ACS-7B Commercial Pilot ACS - Area I Task H
FAA-S-ACS-25 CFI ACS - Area II Task A

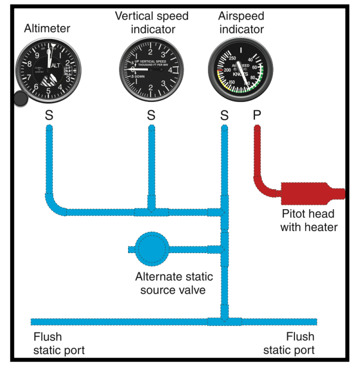
FAA-H-8083-2 Risk Management Handbook
FAA-H-8083-3C Airplane Flying Handbook
FAA-H-8083-9 Aviation Instructors Handbook
FAA-H-8083-25C PHAK Chapter 8: Flight Instruments
FAA-H-8083-25C PHAK Chapter 17: Aeromedical Factors

ERAU Pitot Static Instruments Video
ERAU Gyroscopic Instruments Video

Leading Edge Boot Operation

1. FLIGHT INSTRUMENTS

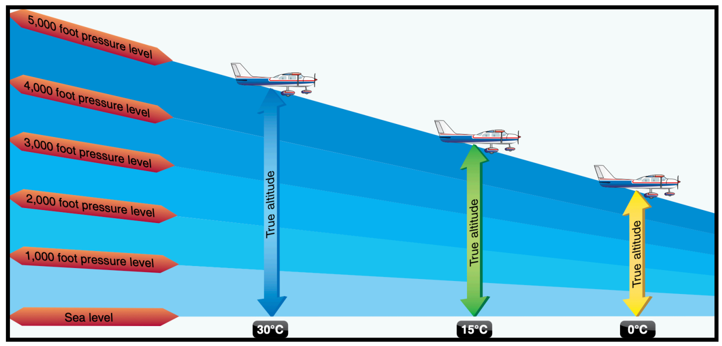
2. PITOT-STATIC SYSTEM AND PRESSURE DRIVEN INSTRUMENTS



Static Pressure – Imagine swimming to the bottom of a deep pool. The deeper you go, the higher the pressure. This is analogous to air; the lower you are, the higher the pressure. As the airplane climbs, the air pressure decreases. We can measure static pressure with a hole perpendicular to the airplanes travel.

Total Pressure – Imagine laying in a hot tub in front of the jet. You can feel this is added pressure. As the airplane flies through the air, it "feels" like a jet is impinging it. We can measure total pressure via a small hole aligned with the airplane's travel.

2.1. Altimeter



- Contain sealed "aneroid" wafers, originally set to 29.92" Hg. The case of the altimeter is vented to static pressure. As status pressure increases or decreases, the wavers contract and expand.
- Errors: blocked static port will freeze indications, since case is no longer venting. Select alternate static

2.2. Vertical Speed Indicator (VSI)

- Contain diaphragm connected to static pressure. Case is vented to a "calibrated leak", which delays changes in static pressure. Until the static pressure and leak pressure equalize, the VSI shows a climb or descent.
- Errors: blocked static port will cause VSI to read zero all the time.

2.3. Airspeed Indicator

- The total pressure is connected to a diaphragm, and the case is connected to static pressure. The differences between these two pressures is "dynamic pressure", which is a function of airspeed and density. A measure of how many molecules of air are caught by the pitot tube.
- Errors: Blocked pitot, open drain - airspeed reads 0. Blocked pitot, blocked drain, airspeed freezes but climbs/descends with altitude. Blocked static, airspeed will read lower after climbing and higher after descending.

3. GYROSCOPIC DRIVEN INSTRUMENTS

Properties of gyroscopes – rigidity in space and precession. Due to angular momentum.

3.1. Attitude Indicator

Property: rigidity in space

Internal gyro spins via suction from a vacuum pump connected to the engine. The attitude indicator rotates in a horizontal plane, and the airplane rolls and pitches around the gyro. Pendulous vanes restore the attitude indicator to straight and level over time. **Explain markings.**

3.2. Heading Indicator / HSI

Property: rigidity in space

Internal gyro spins via suction from a vacuum pump connected to the engine. The heading indicator rotates in a vertical plane but maintains its orientation as the airplane turns around it. These instruments are affected by precession and require reseting eery 15-30 minutes.

3.3. Turn Coordinator

Property: precession

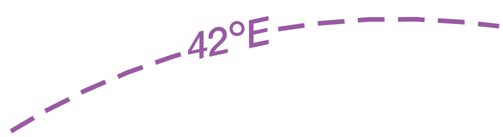
Internal gyro spins via a motor to provide redundancy with the vacuum pump. The gyro rotates in a vertical plane and indicates rate of turn and initial rate of roll. A pilot could be in a bank without turning and the turn coordinator would show level.

4. MAGNETIC COMPASS

4.1. Variation

The Earth's "true" poles are constant. However, the magnetic poles move, albeit slowly. As a result, we have magnetic headings and true headings. We will experience these first hand when we learn about navigation.

When we plan out route on the sectional, we read true headings from the lines of latitude and longitude. However, the airplane detects magnetic headings. The conversion is also depicted on the sectional as isogonic lines. The number is the amount of heading to add or subtract from true. Subtract east variation and add west variation. "East is least, west is best."



4.2. Deviation

Deviation occurs form the airplanes onboard systems interfering with the magnetic compass, and is dependent on airplane heading. Under 23.1547[1] a compass calibration card, which depicts what indicated magnetic heading to fly to achieve an actual magnetic heading is required equipment.

4.3. Turning Errors

UNOS – Undershoot north, overshoot south, by about 30° here in SoCal.

4.4. Acceleration/Deceleration Errors

ANDS – Accelerate North, decelerate South.

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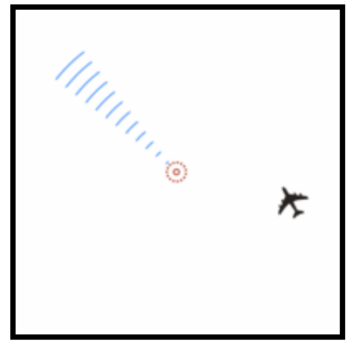
[1] 23.1547 no longer exists. However, per the Tomahawk's TCDS, it must be adhered to.

5. NAVIGATION EQUIPMENT

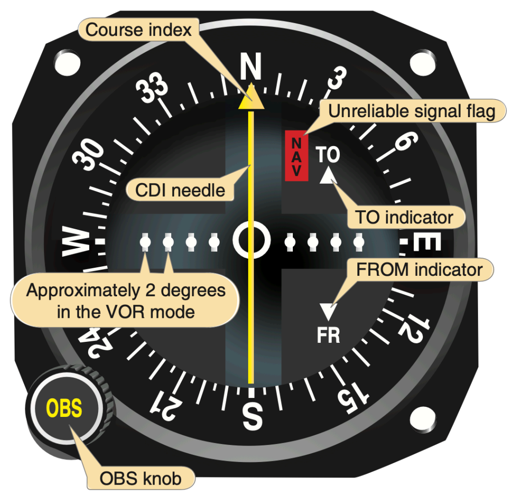
5.1. VHF omnirange (VOR).



Imagine a lighthouse. This particular lighthouse has a rotating light that sweeps through 360°, but every time it passes through north a red light on top flashes for every direction. If we time the interval between the red light and the rotating light, we can determine our direction from the station!

We can capitalize on this even further. In the cockpit, we can set a desired heading to track inboard to or outbound from a VOR station. Headings flying toward the station are known as bearings, and headings flying away from the station are known as radials (radiating outward).

The typical instrument to receive VOR signals is known as a VOR indicator.
Using the knob labeled "OBS" (omni bearing selector), we can select a bearing or radial. If our airplane is directly along this imaginary line, the course deviation indicator (CDI) needle is centered. The further we are off course, the more the needle deflects. Each dot on the face of then indicator represents 2° off course, for a full scale deflection of 10°. This is not the distance, but rather the angle off course. We say "fly toward the needle" – if it is to the left, turn left. The needle shows where the track is relative to our airplane.



If the **bearing to** the station and selected OBS are within 90°, a TO flag will show. If they are greater than 90°, a FROM flag will show. When flying TO a station, always fly the selected course with a TO indication. When flying FROM a station, always fly the selected course with a FROM indication.

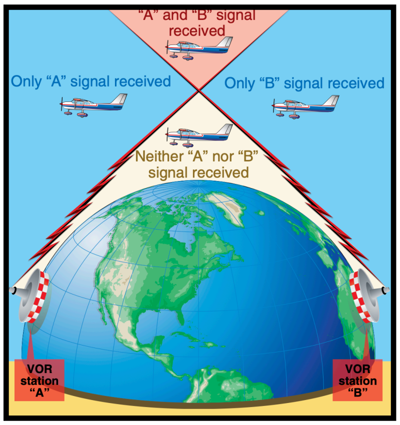
Lets say we want to track northbound to a station. We will select the 360° radial, confirm we have a TO indication, and fly heading 360°. When we pass the station, the TO flag will flip to a FROM flag. This is the expected behavior.

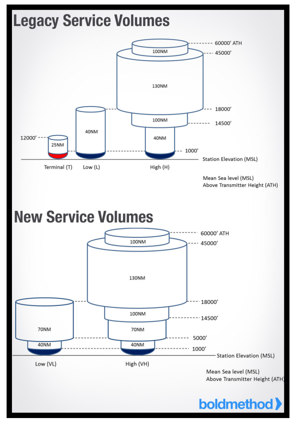
Lets say we accidentally select 180° on the OBS when we are south of the station and wanting to fly toward the station. The indication will read FROM, since it thinks we are trying to track outbound. If we are in fact flying inbound, this is known as **reverse sensing.** The CDI will move in the **opposite** direction as expected. Then we say "pull the needle."

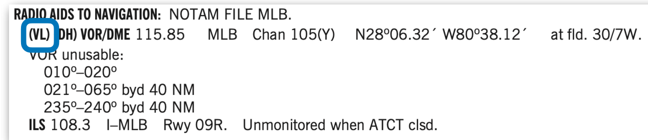
VOR bearings and radials are relative to magnetic north. This is clearly seen on the VFR sectional, where the compass rose does not align with north/south lines of latitude. However, VORs are not recalibrated every time magnetic variation is updated in an area. As time passes, the VOR's bearings and radials become less and less accurate. The variation when the VOR was last calibration is known as declination, and is found in the chart supplement or ForeFlight.

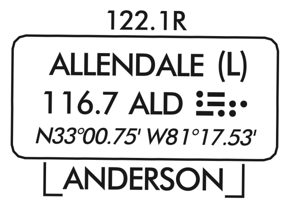
If we plan on using the VOR for IFR navigation, we need to have it inspected per 91.171.

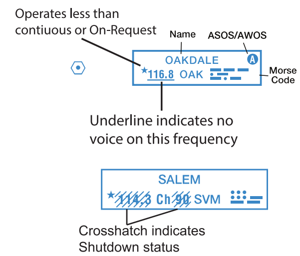
VORs are line of sight navigation equipment. Additionally, they have service volumes shown below. Service volumes are shown on an IFR low chart and the chart supplement.



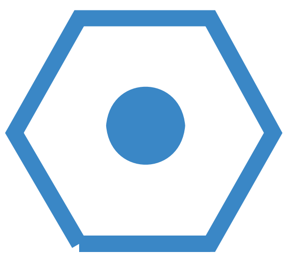


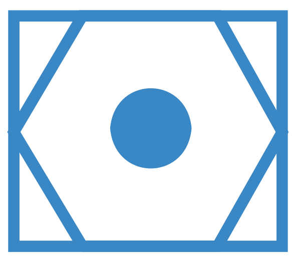


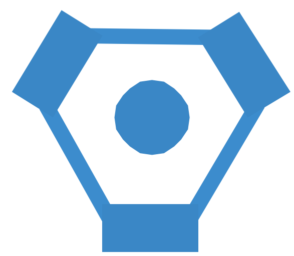




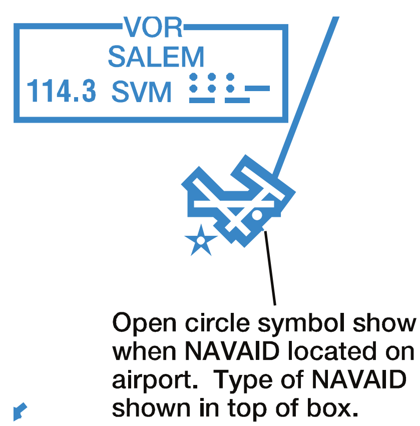
VORs may be colocated with distance measuring equipment (DME) or a Tacan (military version of a DME). These are shown by the symbols below.





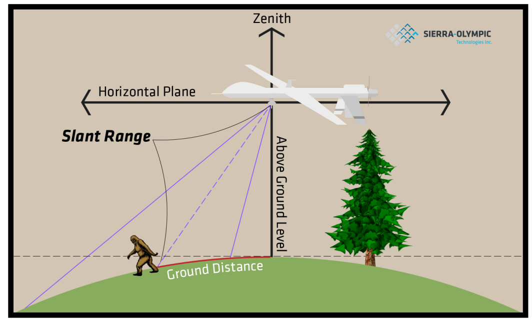


If located on an airport, the located is shown with an open circle. The type of VOR is then depicted at the top of the information box. Sometimes, AWOS, ASOS, or HIWAS may be broadcast over the VOR frequency!



5.2. Distance Measuring Equipment (DME)

VORs give us no distance information. However, if your airplane is equipped with a DME indicator, simply tune the corresponding NAV frequency, and the SLANT RANGE distance will appear. Note that this is not horizontal distance, but point to point distance!

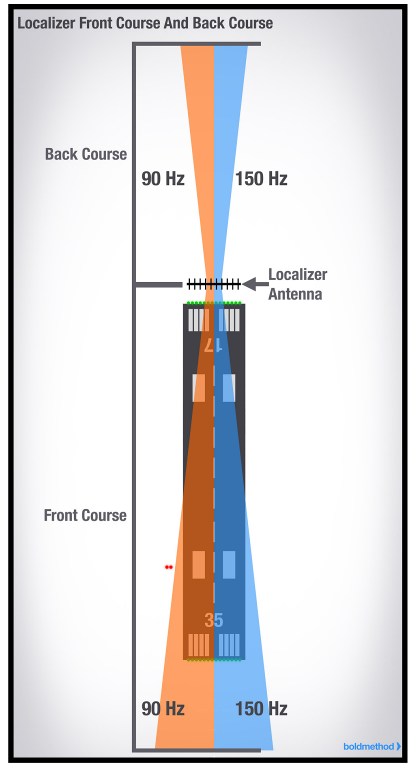


5.3. Instrument Landing System (ILS)

VORs are great for long distance navigation. However, to shoot an approach, we need a higher accuracy solution. The instrument landing system has two components: the localizer and glideslope.

5.3.1. Localizer

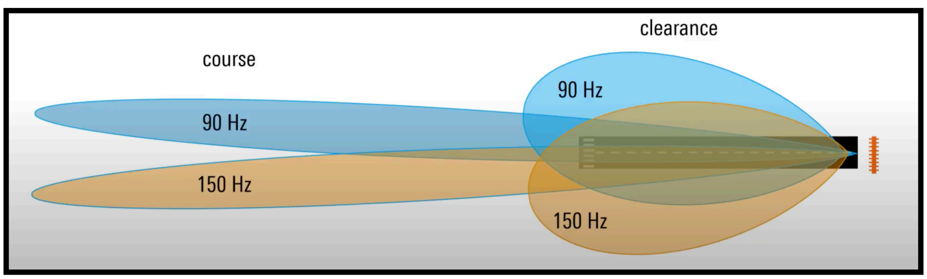
The localizer uses the same VOR indicator to show lateral deviation relative to the runway's extended centerline. Selecting a course with the OBS is useless because the localizer is only designed to function on one course – to and from the runway.



The localizer antenna, located on the far side of the runway, broadcasts two signals on both the approach and departure sides of the runway, shown as the blue and orange shaded regions in the image. These signals are overlapped to form lobes. By comparing the respective strength of each signal, the left or right deviation can be measured.

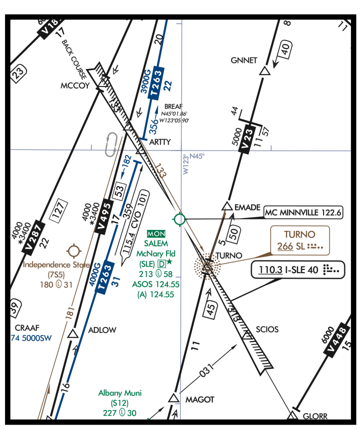
Like the VOR, the deflection is based off angular deviation, not total distance. The localizer then becomes more sensitive at closer distances.

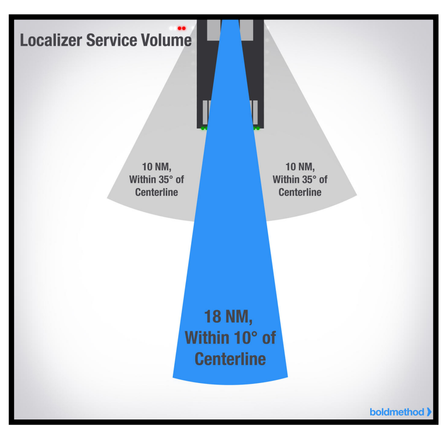
The width of the localizer varies, as it is tuned to be 700ft wide at the runway threshold. A full scale deflection is usually between 2-3° off course.



As mentioned before, the localized signal is radiated on both the approach and departure end of the runway. However, the indictor is expecting the 90 Hz signal on the left side, and the 150 Hz signal on the right. When tracking outbound on the missed approach this is a non-issue. However, some airports capitalize on this and publish a "Localizer Backcourse", which has airplanes fly inbound on what is normally the outbound side. Because the signals are reversed, the airplane reverse senses.

On IFR low charts, the localizer may be shown to provide additional sources to cross-reference. The localizer service volume is shown below.





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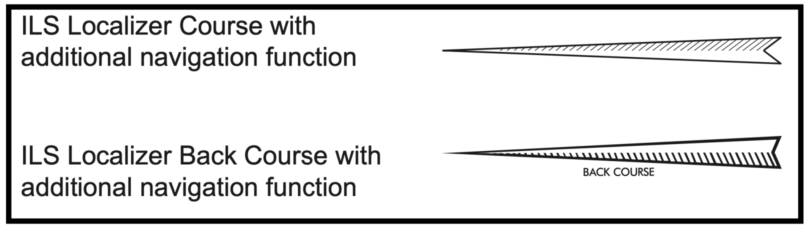
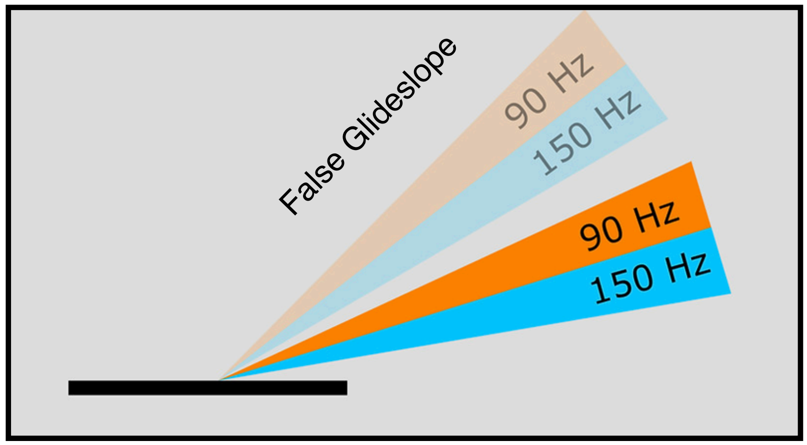
5.3.2. Glideslope

The glideslope functions similarly to a localize on its side. The glideslope antenna array is ~1000 feet ahead of the end of the runway, and emits to lobes that the aircrafts receiver compares to calculate height. The glideslope is 1.4° thick, pointed usually 3° up. Glideslopes are only pointed toward the arrival side of the runway; there is no backcourse.

Due to ground reflections, “false glideslopes” may form above the actual glideslope. Aircraft should ensure they intercept the glideslope from below.

Glideslope sensitivity also increases with proximity. At 10 NM, a full glideslope deflection may represent near 1500 ft. Over the threshold, it drops to a few feet.





5.4. Deprecated Navigation Methods

5.4.1. Marker Beacons

A marker beacon isn't so much a navigation method, but an instantaneous position indicator. When the pilot flies over a marker beacon (typically located along an instrument approach), a light or tone on the audio panel will indicate station crossing.

5.4.2. Non-Directional Beacons (NDBs) and Automatic Direction Finders (ADF)

The NDB is the simplest of navigation methods. The station broadcasts a signal in all directions, and the cockpit indicator (ADF) points in the direction of the station. If the airplane's current heading is aligned with on the outer ring, the ADF indicates the bearing to the station.

5.5. Transponder Altitude Encoding

Mode C transponders require an altitude source to broadcast to ATC. A small computer in the altimeter reads pressure altitude, and electronically sends it to the transponder. Altimeter settings do not affect this reading.

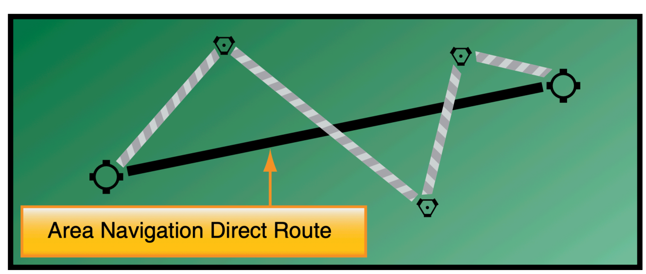
5.6. Global Positioning System (GPS)

Imagine you have four friends. You are lost in the city, so you call your friends for help. Each friend tells you where THEY are and they scream out the current time. You look at your watch, and you compare the time you heard the message to the time you received it. Since you know how fast sound travels, you can determine how far each friend is from you, allowing you to triangulate your position. You need three fiends to triangulate your position, and four friends to also determine the current time.

In reality, the friends are the 31 satellites in the GPS constellation. Each satellite continuously broadcasts its location and the current time that the GPS receiver interprets into a position. This allows for accuracy to 50 ft.

If five satellites are in view, the GPS can perform receiver autonomous integrity monitoring (RAIM), which would identify one of the satellites is corrupt or broken. If six satellites are in view, the corrupt satellite can be removed from the GPS solution.

GPS technology has improved with the advent of WAAS. WAAS, or wide area augmentation system, is a network of ground stations that receive GPS data and relay it to a master station, where the data is processed and a correction is uploaded back to the satellites and aircraft in the coverage area. The accuracy is improved to 10 ft for WAAS enabled receivers. Having WAAS capability negates the need for RAIM monitoring. The Garmin GPS 175 installed in our Tomahawk is a WAAS GPS receiver.



GPS allows for "area navigation," or RNAV. Rather than flying from VOR to VOR, we can display a course from anywhere to anywhere on out CDI, just like a VOR.

6. RADAR/LIGHTNING DETECTION SYSTEMS

7. OTHER INFLIGHT WEATHER SYSTEMS.